Code-along 01

FirstName LastName

Packages

Install Packages

Click the green arrow icon to run the following code chunks.

These chunks will not run (on purpose!) when rendered because `eval: false`.

Install the `tidyverse()` pacakge.

```
install.packages("tidyverse")
```

Install the `tinytex()` pacakge. This package is needed to Render Quarto documents as .pdf.

```
install.packages('tinytex')
```

Then, use `tinytex` package to install a part of the package.

```
tinytex::install_tinytex()
```

Install the `gssr()` package, NOT on CRAN.

```
# Install 'gssr' from 'ropensci' universe
install.packages('gssr', repos =
  c('https://kjhealy.r-universe.dev', 'https://cloud.r-project.org'))
```

Install the `gssrdoc()` package, NOT on CRAN.

```r
# Install 'gssrdoc' from 'ropensci' universe
install.packages('gssrdoc', repos =
  c('https://kjhealy.r-universe.dev', 'https://cloud.r-project.org'))
```

Load the packages

We'll use the following packages:

- `tidyverse()` (data wrangling)
- `gssr()` (U.S. General Social Survey data)
- `gssrdoc()` (GSS documentation)

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```r
library(gssr)
```

```
Package loaded. To attach the GSS data, type data(gss_all) at the console.
For the panel data and documentation, type e.g. data(gss_panel08_long) and data(gss_panel_doc
For help on a specific GSS variable, type ?varname at the console.
```

```r
library(gssrdoc)
```

Environment

```r
# software documentation
sessionInfo()
```

```
R version 4.5.1 (2025-06-13 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26100)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_Canada.utf8  LC_CTYPE=English_Canada.utf8
[3] LC_MONETARY=English_Canada.utf8 LC_NUMERIC=C
[5] LC_TIME=English_Canada.utf8

time zone: America/Toronto
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] gssrdoc_0.7.0   gssr_0.7        lubridate_1.9.4 forcats_1.0.0
 [5] stringr_1.5.1   dplyr_1.1.4     purrr_1.0.4     readr_2.1.5
 [9] tidyr_1.3.1     tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.5.1     tidyselect_1.2.1
 [5] scales_1.4.0       yaml_2.3.10        fastmap_1.2.0      R6_2.6.1
 [9] generics_0.1.4     curl_6.3.0         knitr_1.50         pillar_1.11.0
[13] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        stringi_1.8.7
[17] xfun_0.52          fs_1.6.6           timechange_0.3.0   cli_3.6.5
[21] withr_3.0.2        magrittr_2.0.3     digest_0.6.37      grid_4.5.1
[25] rstudioapi_0.17.1  haven_2.5.5        hms_1.1.3          lifecycle_1.0.4
[29] vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0         farver_2.1.2
[33] rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3    htmltools_0.5.8.1
```

Meet your data

We're going to use data from the U.S. General Social Survey (GSS).

Let's load your data.

```r
# Get the data only for the 2024 survey respondents
gss24 <- gss_get_yr(2024)
```

```
Fetching: https://gss.norc.org/documents/stata/2024_stata.zip
```

```
# look at the dataframe
gss24
```

```
# A tibble: 3,309 x 639
   year       id wrkstat hrs1        hrs2        evwork      marital martype
   <dbl+l> <dbl> <dbl+l> <dbl+lbl>   <dbl+lbl>   <dbl+lbl>   <dbl+l> <dbl+lbl>
 1 2024        1 1 [wor~    43       NA(i) [iap] NA(i) [iap] 5 [nev~ NA(i) [iap]
 2 2024        2 5 [ret~ NA(i) [iap] NA(i) [iap]     1 [yes] 5 [nev~ NA(i) [iap]
 3 2024        3 5 [ret~ NA(i) [iap] NA(i) [iap]     1 [yes] 1 [mar~     1 [mar~
 4 2024        4 2 [wor~    20       NA(i) [iap] NA(i) [iap] 5 [nev~ NA(i) [iap]
 5 2024        5 5 [ret~ NA(i) [iap] NA(i) [iap]     1 [yes] 3 [div~ NA(i) [iap]
 6 2024        6 4 [une~ NA(i) [iap] NA(i) [iap] NA(i) [iap] 1 [mar~     1 [mar~
 7 2024        7 1 [wor~    80       NA(i) [iap] NA(i) [iap] 1 [mar~     1 [mar~
 8 2024        8 6 [in ~ NA(i) [iap] NA(i) [iap]     1 [yes] 3 [div~ NA(i) [iap]
 9 2024        9 1 [wor~    50       NA(i) [iap] NA(i) [iap] 1 [mar~ NA(i) [iap]
10 2024       10 4 [une~ NA(i) [iap] NA(i) [iap] NA(i) [iap] 5 [nev~ NA(i) [iap]
# i 3,299 more rows
# i 631 more variables: divorce <dbl+lbl>, widowed <dbl+lbl>,
#   spwrksta <dbl+lbl>, sphrs1 <dbl+lbl>, sphrs2 <dbl+lbl>, spevwork <dbl+lbl>,
#   cowrksta <dbl+lbl>, coevwork <dbl+lbl>, cohrs1 <dbl+lbl>, cohrs2 <dbl+lbl>,
#   sibs <dbl+lbl>, childs <dbl+lbl>, age <dbl+lbl>, educ <dbl+lbl>,
#   speduc <dbl+lbl>, coeduc <dbl+lbl>, codeg <dbl+lbl>, degree <dbl+lbl>,
#   padeg <dbl+lbl>, madeg <dbl+lbl>, spdeg <dbl+lbl>, sex <dbl+lbl>, ...
```

A "tibble" is another name for "tidy dataset," meaning that the data is organized in structured, clear rows and columns. "(3,309 × 639)" means the dataset contains 3,309 rows and 639 columns. In social sciences, rows are commonly referred to as "observations" and columns as "variables." In our case, there are 3,309 observations (e.g., respondents) and 639 variables.

With your mouse, go to the environment panel (upper-right) and click on the `gss24` object. It pops up and you can browse through it.

This is often a good idea to get a first feel for the data, but only if your dataset is relatively small.

Variables

The GSS documentation is also available online in .pdf form.

The .pdfs will be useful for general overviews.

For specific variable information, it will be helpful to use the documentation you'll load into RStudio.

For information about a specific GSS variable, type `?varname` at the console.

In the output pane, the Help tab will show the variable documentation.

Example:

```
# copy/paste this code in the Console; don't run in Quarto
?meovrwrk
```

Let's look at a frequency table of this variable.

(NOTE: The variable names are case sensitive. In this dataset, all variables are lowercase.)

```
table(gss24$meovrwrk)
```

```

  1   2   3   4   5 
195 787 611 481  89 
```

195 respondents were coded as 1 on this variable. What does that mean? (Look at the help page.)

Change the code to show the variable `fefam`.

Add text here that interprets the results for the `2` value for this variable.

```
# Replace meovrwrk with fefam
table(gss24$meovrwrk)
```

```

  1   2   3   4   5 
195 787 611 481  89 
```

Quarto: Render

Finally, `Render` your code-along-01 and view the results.